CSET485 – AI and Society

Stress-Testing a Predictive Model on the COMPAS Recidivism Dataset

Assignment #1 (10 Marks): Build, Stress-Test and Defend

Student	Sakshi Arora
Roll number	0450
Assigned dataset	COMPAS Recidivism Risk Score Dataset (roll number ends in 0)
Random seed / split	random_state = 450 (last three digits of the roll number); 5,049 training rows (70%), 2,165 test rows (30%)
Data source	ProPublica compas-scores-two-years.csv (Broward County, Florida)
Companion code	Jupyter notebook CSET485_Assignment1_COMPAS.ipynb; Appendix C explains how to regenerate every number in this report

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Executive summary

This report stress-tests risk models built on the COMPAS two-year recidivism file (n = 7,214; seed 450; 5,049 training and 2,165 test rows). A linear model of the COMPAS decile score (Model A) and a logistic model of actual two-year recidivism (Model B; test accuracy 67.9%, AUC 0.734) disagree on 296 test defendants (13.67%), almost all of them flagged by Model A only. Four stress tests follow. (1) No genuine sign reversal exists; the clearest subgroup difference is a stronger prior-record effect for women (1.024 versus 0.666 for men), which shrinks once priors are log-transformed. (2) Omitting priors_count turns the race coefficient from 0.098 (not significant) into 0.302 ($p < 0.001$). (3) 90 confident errors (4.16% of the test set) are almost exactly as many as the model's own probabilities predict. (4) Across thresholds 0.05 to 0.95, demographic parity and equalized odds rise and fall together ($r = 0.998$); the real cost of parity is accuracy and predictive parity. I recommend conditional, advisory-only use at $t = 0.45$ (DP 0.507, EO 0.478, PP 0.920, accuracy 66.8%), with the limitations documented below.

1 Introduction and objectives

Predictive models used in high-stakes decisions can look statistically sound while concealing bias, hidden confounding and fairness trade-offs. This assignment asks for a model to be built, deliberately attacked, and then defended in writing using only numbers the student computed. The objectives are: (i) fit a linear (Model A) and a logistic (Model B) model on the assigned dataset and compare the high-risk labels they produce; (ii) run four stress tests: a Simpson's-paradox hunt, an omitted-variable experiment, an adversarial subset of confident errors, and a fairness-threshold sweep; (iii) answer eight analytical questions; and (iv) reflect on whether a policy-maker can responsibly be given “just one number”. My roll number (0450) ends in 0, so the assigned dataset is the COMPAS Recidivism Risk Score file, and the random seed and split are 450 and 70/30. All results come from one notebook that regenerates every table, figure and number in this document.

2 Dataset description

The data are ProPublica’s *compas-scores-two-years.csv* (7,214 defendants, 53 columns) from Broward County, Florida, downloaded from the URL in the assignment (SHA-256 recorded in Appendix C). Defendant IDs are unique; *decile_score* ranges from 1 to 10. **Target-variable check.** The brief names a binary outcome *is_recidivist*, but no such column exists. Two candidates do: *two_year_recid* and *is_recid*, which disagree on 220 people who reoffended after the two-year window. I therefore define *is_recidivist := two_year_recid* because it alone has a fixed two-year follow-up; its base rate is 45.1% (versus 48.1% for *is_recid*). A sensitivity check with *is_recid* (Appendix B) changes AUC by only 0.005. Features are age, priors_count, three juvenile-record counts, sex, charge degree (felony/misdemeanour) and race. The recidivism rate differs by group (Table 1), as do prior records and age, which matters for every fairness result that follows.

Table 1. Dataset summary by demographic group (all 7,214 rows).

Group	n	Share of data	Two-year recidivism rate	Mean COMPAS decile	Mean priors	Mean age
African-American	3,696	51.2%	51.4%	5.37	4.44	32.7
Caucasian	2,454	34.0%	39.4%	3.73	2.59	37.7
Hispanic	637	8.8%	36.4%	3.46	2.25	35.5
Other (Asian, Native American, Other; pooled)	427	5.9%	35.6%	3.08	2.02	35.1
Male	5,819	80.7%	47.3%	4.59	3.76	34.8
Female	1,395	19.3%	35.7%	4.17	2.28	34.9
All defendants	**7,214**	**100.0%**	**45.1%**	**4.51**	**3.47**	**34.8**

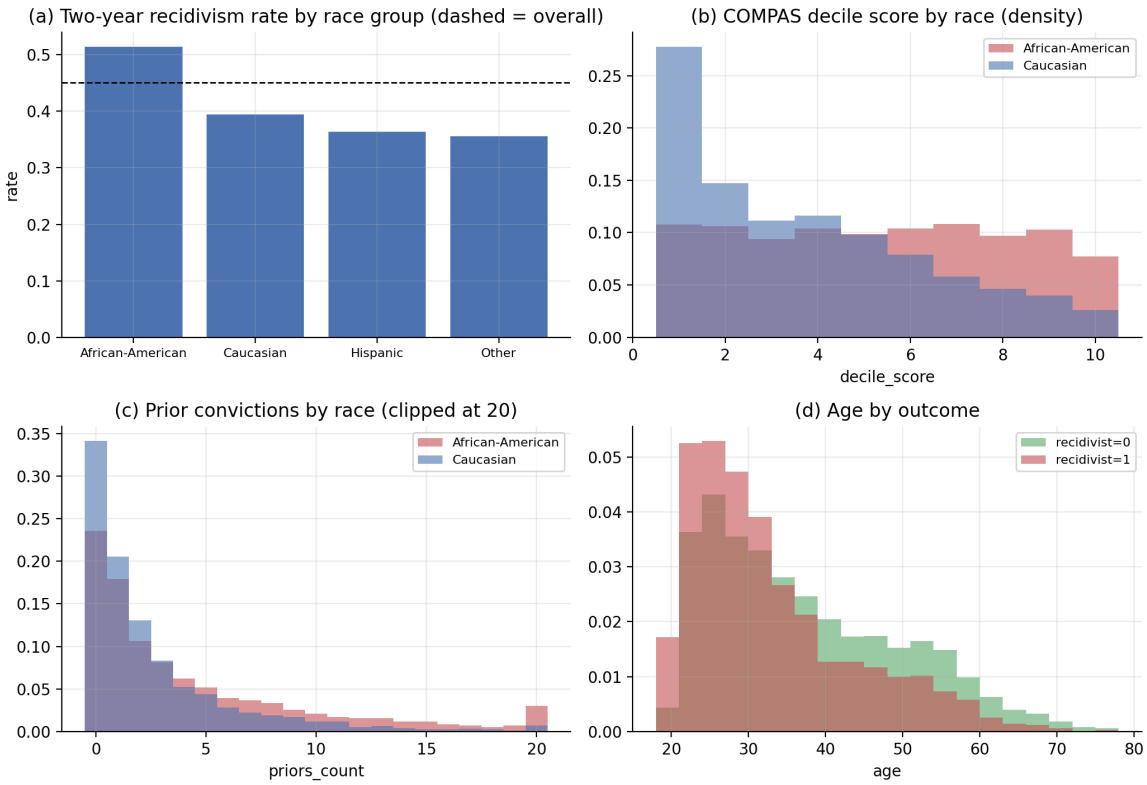


Figure 1. Exploratory analysis: (a) recidivism rate by race group, (b) COMPAS decile score, (c) prior convictions and (d) age by outcome.

3 Data preprocessing methodology

Each preprocessing decision is documented below, as the assignment requires. No outcome-derived or COMPAS-output column is used as a predictor, and every fitted transformation (the scaler) is estimated on the training set only.

Table 2. Preprocessing decisions.

Issue	Decision and justification
Missing values	22 of the 53 columns contain missing values (dates, charge descriptions, follow-up and violent-recidivism fields). None of them is used as a predictor, so **no imputation** is needed and **no row is dropped** (n = 7,214); the ten model features have zero missing values (asserted in the notebook).
Leakage control	Excluded as predictors: *two_year_recid*, *is_recid*, every *r_**, *vr_** and *violent_recid* column (post-screening outcomes); *decile_score*, *score_text* and *v_** (COMPAS's own output, which is Model A's target and a proxy for the outcome); identifiers and dates. The scaler is fitted on the training set only.
Categorical encoding	*sex* → *male* (1/0); *c_charge_degree* → *felony* (F = 1, M = 0); *race* → one-hot with Caucasian as reference. Asian (n = 32), Native American (n = 18) and Other (n = 377) are pooled into *race_Other* (n = 427) because the two smallest groups are too small to estimate stably. *age_cat* is dropped (redundant with *age*).
Feature scaling	The five numeric features (age, priors_count, juv_fel_count, juv_misd_count, juv_other_count) are z-scored with a *StandardScaler* fitted on the 5,049 training rows and applied unchanged to the test set. Coefficients are therefore per 1 SD of the training distribution; dummy variables are unscaled.
Outliers	IQR-rule outliers are counted in the next table. All values are plausible records (age ≤ 96, priors ≤ 38) and the long right tail of priors_count is the high-risk population, so **values are retained**. A check that winsorises priors_count at the training 99th percentile (22) changes test AUC by 0.001 (Appendix B).
Target definition	*is_recidivist* := *two_year_recid* (see Section 2). Model A's target is *decile_score* (1–10).

Table 3. Outlier summary for the numeric features (all retained).

Feature	Min	Median	Mean	99th pct	Max	IQR outliers	% of rows
age	18	31	34.82	67	96	61	0.8%
priors_count	0	2	3.47	23	38	455	6.3%
juv_fel_count	0	0	0.07	2	20	282	3.9%
juv_misd_count	0	0	0.09	2	13	415	5.8%
juv_other_count	0	0	0.11	2	17	523	7.2%

4 Part 1 – Model development, evaluation and mismatch analysis

Models. Model A is ordinary least squares on *decile_score*; Model B is unpenalised logistic regression on *is_recidivist*. Both use the same ten features and the same split, and were cross-checked against scikit-learn (maximum prediction difference 3.6e-14 for A and 4.1e-07 for B). On the test set Model A reaches $R^2 = 0.422$ (RMSE 2.173 deciles, MAE 1.754); its predictions range from −1.48 to 15.67, i.e. outside the valid 1–10 scale, a known weakness of linear models. Model B reaches accuracy 67.9%, AUC 0.734, precision 0.677 and recall 0.561 at the 0.5 threshold; the majority-class baseline is 54.6%, so the model adds 13.3 percentage points. **Threshold comparison.** Model A's test median prediction is 4.592; flagging predictions ≥ that median labels 1,084 defendants (50.07%) high risk. Flagging Model B probabilities ≥ 0.5 labels 814 (37.60%). The two labellings disagree on **296 defendants (13.67%)**: 283 are high risk under A only and 13 under B only. Table 7 distinguishes this *agreement* from *accuracy against actual outcomes*. The reasons for the disagreement are analysed in the answer to Question 1.

Table 4. Model coefficients on the training set (numeric predictors standardised; Model A in decile points, Model B in log-odds).

Predictor	Model A (OLS): β (SE)	p	Model B (logit): β	95% CI	p	Odds ratio
const	4.036 (0.089)	<0.001	−0.640	[−0.820, −0.461]	<0.001	
age	−1.192 (0.033)	<0.001	−0.526	[−0.596, −0.455]	<0.001	0.591
priors_count	1.233 (0.034)	<0.001	0.701	[0.621, 0.782]	<0.001	2.017
juv_fel_count	0.178 (0.031)	<0.001	0.054	[−0.036, 0.143]	0.238	1.055
juv_misd_count	0.133 (0.033)	<0.001	−0.018	[−0.087, 0.051]	0.612	0.982
juv_other_count	0.140 (0.032)	<0.001	0.049	[−0.021, 0.119]	0.170	1.050
male	−0.052 (0.078)	0.502	0.319	[0.164, 0.475]	<0.001	1.376
felony	0.387 (0.065)	<0.001	0.208	[0.081, 0.335]	0.001	1.231
race_African_American	0.587 (0.070)	<0.001	0.098	[−0.038, 0.235]	0.158	1.103
race_Hispanic	−0.401 (0.113)	<0.001	−0.088	[−0.313, 0.137]	0.441	0.915
race_Other	−0.796 (0.135)	<0.001	−0.180	[−0.449, 0.089]	0.190	0.835

Table 5. Test-set performance of Models A and B.

Model	Metric (test set unless stated)	Value
A (OLS)	R^2 (train / test)	0.431 / 0.422
	RMSE / MAE (deciles)	2.173 / 1.754
	Median / min / max prediction	4.592 / −1.48 / 15.67
B (logit)	Accuracy at 0.5 (majority baseline)	0.679 (0.546)
	AUC / Brier score / log-loss	0.734 / 0.208 / 0.606
	Precision / recall / F1	0.677 / 0.561 / 0.613
	McFadden pseudo-R^2 (train)	0.105

Table 6. Confusion matrix of the two thresholding approaches (rows: Model A at the median 4.592; columns: Model B at 0.5). Mismatched cases are the off-diagonal cells: 13 + 283 = **296 of 2,165 (13.67%)**.

	Model B: low risk (p < 0.5)	Model B: high risk (p ≥ 0.5)	Total
Model A: low risk (< median)	1,068	13	1,081
Model A: high risk (≥ median)	283	801	1,084
Total	**1,351**	**814**	**2,165**

Table 7. Agreement is not accuracy: each flag scored against the *actual* outcome. Both models are correct together on 1,301 defendants and wrong together on 568; on the 296 disagreements Model B is right 169 times and Model A 127 times.

Flag	Flag rate	Accuracy	Precision	Recall	FPR	TP	FP	TN	FN
Model A (≥ median)	50.1%	66.0%	61.3%	67.7%	35.4%	665	419	763	318
Model B (p ≥ 0.5)	37.6%	67.9%	67.7%	56.1%	22.3%	551	263	919	432

The explanation of *why* the two approaches flag different people is given in Part 3, Question 1.

5 Part 2.1 – Simpson's-paradox hunt

Design. Inference models were fitted on all 7,214 rows (full) and on each subgroup with identical features (the split variable omitted). Differences are tested with $z = (\beta_A - \beta_B)/\sqrt{(se_A^2 + se_B^2)}$. Because I was searching for a reversal, I first scanned three demographic splits (race, sex, age ≤ 30 vs > 30) over every predictor: 26 comparisons, so the Bonferroni threshold is $p < 0.0019$. **Finding 1: no genuine reversal.** 5 of the 26 coefficients change sign between subgroups, but in none of them are both subgroup coefficients significantly non-zero (0 of 5), and none is significant after correction; they are sampling noise around zero. For example, the race coefficient is 0.111 [−0.017, 0.238] for men and −0.231 [−0.491, 0.029] for women (difference p = 0.021, Bonferroni p = 0.54); I report it as an exploratory observation, not a finding. **Finding 2: a significantly stronger effect.** The priors_count effect is 1.024 [0.813, 1.235] for women (B, n = 1,395) versus 0.666 [0.594, 0.738] for men (A, n = 5,819); full data 0.721. The intervals do not overlap (z = 3.15, p = 0.0016; Bonferroni p = 0.043). It is the only sex-split difference that survives correction (age-split differences are partly mechanical because age is both the splitter and a predictor). **Diagnostic.** Entering priors as log(1 + priors) shrinks the gap to 0.666 (men) versus 0.811 (women), p = 0.075, so much of the heterogeneity reflects functional form: women have fewer priors (mean 2.28 versus 3.76). The race split, which matters most for Section 2.4, yields priors_count coefficients of 0.664 (African-American) versus 0.856 (Caucasian), nominal p = 0.021 but Bonferroni p = 0.55: suggestive, not established (Table 9).

Table 8. Primary split: sex (A = male, n = 5,819; B = female, n = 1,395); logistic coefficients per 1 SD, full data n = 7,214. Bonferroni over 26 tests. The priors_count row is the finding.

Predictor	Full β	Male (A) β [95% CI]	Female (B) β [95% CI]	z (A−B)	p	p (Bonf.)
age	−0.527	−0.542 [−0.607, −0.476]	−0.468 [−0.610, −0.327]	−0.92	0.359	1.000
priors_count	**0.721**	**0.666 [0.594, 0.738]**	**1.024 [0.813, 1.235]**	**−3.15**	**0.002**	**0.043**
juv_fel_count	0.081	0.078 [0.003, 0.154]	0.046 [−0.351, 0.442]	0.16	0.875	1.000
juv_misd_count	0.010	0.012 [−0.050, 0.074]	−0.026 [−0.211, 0.159]	0.39	0.699	1.000
juv_other_count	0.100	0.093 [0.027, 0.159]	0.065 [−0.121, 0.251]	0.28	0.780	1.000
felony	0.180	0.123 [0.004, 0.242]	0.331 [0.083, 0.580]	−1.48	0.138	1.000
race_African_American	0.049	0.111 [−0.017, 0.238]	−0.231 [−0.491, 0.029]	2.31	0.021	0.538
race_Hispanic	−0.176	−0.196 [−0.406, 0.015]	−0.142 [−0.622, 0.338]	−0.20	0.842	1.000
race_Other	−0.188	−0.163 [−0.410, 0.084]	−0.386 [−0.963, 0.191]	0.69	0.487	1.000

Table 9. Secondary split: race (A = African-American, n = 3,696; B = Caucasian, n = 2,454); race indicators omitted from all three models. No difference survives correction.

Predictor	Full β	Afr.-Amer. (A) β [95% CI]	Caucasian (B) β [95% CI]	z (A−B)	p	p (Bonf.)
age	−0.534	−0.620 [−0.711, −0.528]	−0.435 [−0.526, −0.344]	−2.80	0.005	0.132
priors_count	0.722	0.664 [0.579, 0.749]	0.856 [0.717, 0.995]	−2.31	0.021	0.545
juv_fel_count	0.077	0.061 [−0.020, 0.142]	0.047 [−0.134, 0.229]	0.14	0.891	1.000
juv_misd_count	0.009	−0.031 [−0.097, 0.035]	0.083 [−0.065, 0.230]	−1.37	0.169	1.000
juv_other_count	0.095	0.177 [0.084, 0.270]	0.009 [−0.087, 0.105]	2.46	0.014	0.365
male	0.287	0.456 [0.271, 0.640]	0.115 [−0.093, 0.323]	2.40	0.016	0.421
felony	0.177	0.134 [−0.017, 0.285]	0.160 [−0.020, 0.341]	−0.22	0.829	1.000

Fig. 2 Logistic coefficients: full data vs subgroups

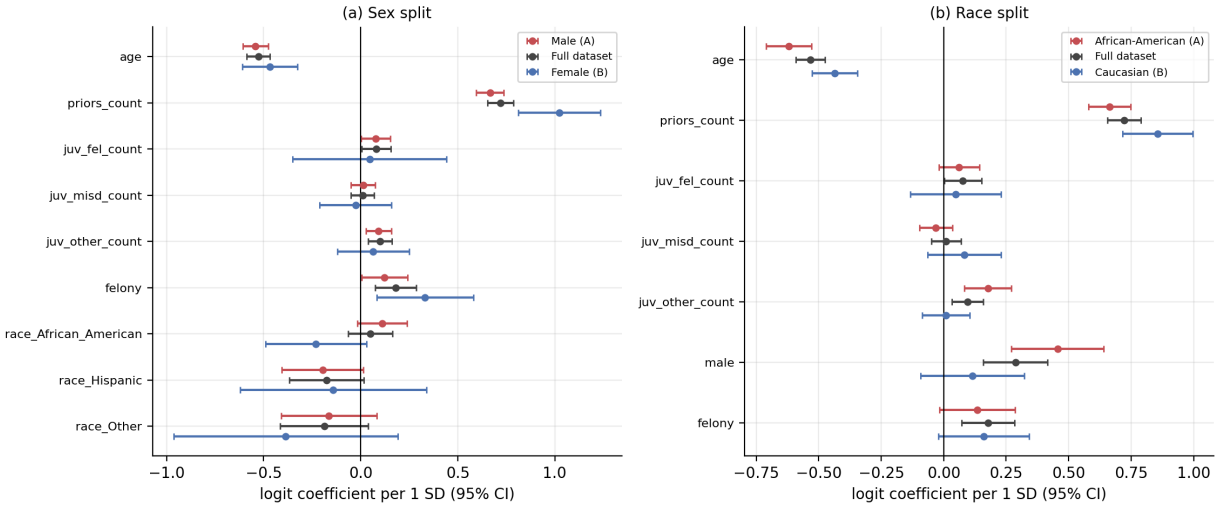


Figure 2. Logistic coefficients with 95% confidence intervals for the full data and each subgroup.

6 Part 2.2 – Omitted-variable bias

Choice of confounder: priors_count. It is the strongest predictor of recidivism (full-model β = 0.701, training-set correlation with the outcome r = 0.27) and is correlated with the other predictors, including race_African_American (r = 0.19), juv_misd_count (r = 0.22) and age (r = 0.17), so removing it should push its explanatory power onto them. It is also shaped by policing intensity, which is why it is a contestable variable. The full model is Model B (training set); the reduced model drops only priors_count. **Formula:** $\%\Delta = (\beta_{reduced} - \beta_{full}) / |\beta_{full}| \times 100$; because this explodes when $\beta_{full} \approx 0$, Table 10 also reports the absolute change and the change in units of the full model's standard error. **Results.** The largest percentage change is juv_misd_count (−0.018 → 0.109, +708%, a sign flip), but its baseline is almost zero, so this is a numerical artefact. The largest absolute and standardised shift is age (−0.526 → −0.312, +40.6%, 5.9 standard errors). The most consequential is race_African_American: 0.098 (p = 0.158) → 0.302 (p < 0.001), +207.7%. Four coefficients (the three juvenile counts and race_African_American) turn from non-significant to significant and none loses significance. Test AUC falls from 0.734 to 0.664. **Caveats.** Logistic coefficients are non-collapsible (they change when covariates are dropped even without confounding), and priors may be a mediator rather than a confounder; the table shows how fragile coefficient interpretations are, not the causal effect of any variable.

Table 10. Coefficients before (full model = Model B) and after dropping priors_count, training set. %Δ = (β_reduced − β_full)/|β_full| × 100; Δ in SE = absolute change divided by the full model's standard error.

Predictor	β full (p)	β reduced (p)	Abs. Δ	% change	Δ in SE	Note
age	−0.526 (<0.001)	−0.312 (<0.001)	0.214	+40.6%	5.92	
juv_fel_count	0.054 (0.238)	0.184 (<0.001)	0.130	+241.3%	2.85	n.s. → sig.
juv_misd_count	−0.018 (0.612)	0.109 (0.002)	0.127	+708.4%	3.60	sign flip; n.s. → sig.
juv_other_count	0.049 (0.170)	0.104 (0.005)	0.055	+111.4%	1.53	n.s. → sig.
male	0.319 (<0.001)	0.434 (<0.001)	0.114	+35.8%	1.45	
felony	0.208 (0.001)	0.354 (<0.001)	0.147	+70.6%	2.27	
race_African_American	0.098 (0.158)	0.302 (<0.001)	0.204	+207.7%	2.93	n.s. → sig.
race_Hispanic	−0.088 (0.441)	−0.124 (0.265)	−0.035	−39.8%	−0.31	
race_Other	−0.180 (0.190)	−0.239 (0.075)	−0.059	−32.6%	−0.43	

Fig. 3 Coefficients before and after omitting priors_count

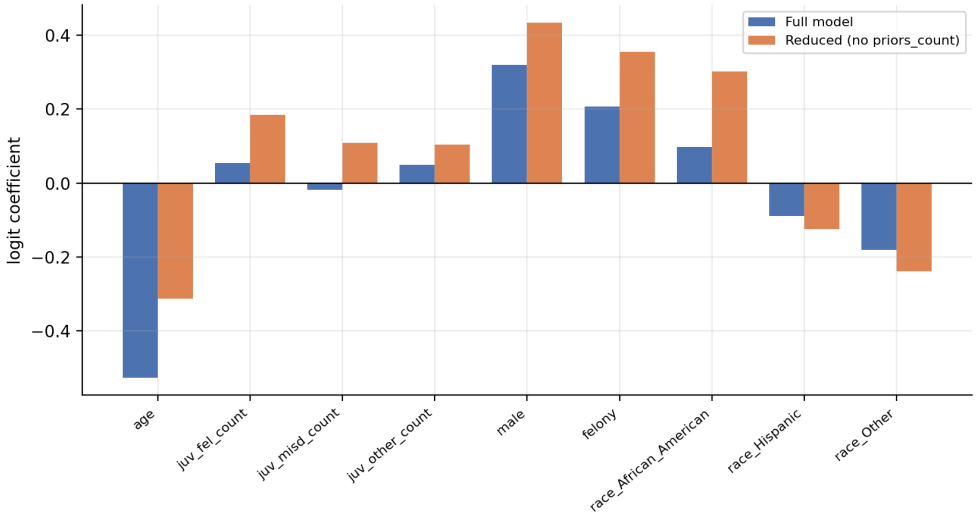


Figure 3. Coefficients of the full and reduced models.

7 Part 2.3 – Adversarial subset construction

Extraction. On the 2,165 test rows there are 35 false positives ($P \geq 0.75$, actual 0) and 55 false negatives ($P \leq 0.25$, actual 1): **90 confident errors, 4.16% of the test set**. Because more than 50 qualify, 50 were drawn at random (*random_state* = 450): 21 false positives and 29 false negatives. **Profile.** Compared with the full test set, the subset contains **no one under 25** (0% versus 20.0%), is much older (median age 47.5 versus 31; 56% aged 45+ versus 21.2%), and has a different gender mix (72% male versus 80.0%). It is two mirror-image clusters rather than one: all 21 false positives have ≥ 5 priors (median 18; 81% African-American, 95% male), whereas the 29 false negatives have none with ≥ 5 priors (76% have at most one, 83% are 45 or older, 17% African-American). **Why the model fails confidently.** (a) *Calibrated uncertainty:* among the 180 test defendants with $P \geq 0.75$ the observed recidivism rate is 80.6% (mean predicted 84.5%), and among the 331 with $P \leq 0.25$ it is 16.6% (predicted 18.2%). A calibrated model should therefore make about 28.0 confident false positives and 60.4 confident false negatives (88.4 in total); the observed 35 and 55 (90) are close, so most of these errors are irreducible outcome noise. (b) *Ignored non-linearity:* adding squared and interaction terms improves fit significantly (LR $\chi^2 = 81.8$, df = 6, $p < 0.001$) with priors$^2 = -0.086$ (diminishing returns) and age$^2 = 0.196$ (older defendants are not as safe as a straight line implies), but test AUC moves only from 0.734 to 0.739 and the count of confident errors does not fall (90 → 114). Misspecification contributes, yet it does not explain most of the failures.

Table 11. Profile of the 50-row adversarial subset compared with the full test set (n = 2,165); FP = false positive, FN = false negative; the last column covers all 90 qualifying errors.

Characteristic	Full test set	Subset (50)	of which FP	of which FN	All errors (90)
n	2165	50	21	29	90
Age, mean	34.7	46.0	38.5	51.3	46.7
Age, median	31.0	47.5	37.0	50.0	48.5
% under 25	20.0	0.0	0.0	0.0	1.1
% aged 45+	21.2	56.0	19.0	82.8	60.0
% male	80.0	72.0	95.2	55.2	77.8
% African-American	51.5	44.0	81.0	17.2	42.2
% Caucasian	34.5	48.0	14.3	72.4	47.8
% Hispanic	8.1	6.0	4.8	6.9	6.7
% Other	5.9	2.0	0.0	3.4	3.3
Priors, mean	3.56	8.22	18.43	0.83	7.07
Priors, median	2.0	2.0	18.0	1.0	2.0
% with 5+ priors	26.6	42.0	100.0	0.0	40.0
% with 0–1 priors	47.2	44.0	0.0	75.9	45.6
% any juvenile record	14.7	12.0	28.6	0.0	11.1
% felony charge	65.6	58.0	81.0	41.4	55.6
% actually reoffended	45.4	58.0	0.0	100.0	61.1
Mean predicted P	0.455	0.468	0.856	0.187	0.439

Table 12. Prevalence of each trait in the subset versus the remaining 2,115 test rows, with Fisher exact p-values (24 tests, uncorrected; read as descriptive).

Trait	Rest of test set	All 50: % (p)	FP only: % (p)	FN only: % (p)
age < 25	20.5%	0.0% (<0.001)	0.0% (0.013)	0.0% (0.002)
age ≥ 45	20.3%	56.0% (<0.001)	19.0% (1.000)	82.8% (<0.001)
male	80.2%	72.0% (0.153)	95.2% (0.100)	55.2% (0.004)
African-American	51.7%	44.0% (0.317)	81.0% (0.008)	17.2% (<0.001)
priors ≥ 5	26.2%	42.0% (0.022)	100.0% (<0.001)	0.0% (<0.001)
priors ≤ 1	47.3%	44.0% (0.670)	0.0% (<0.001)	75.9% (0.002)
any juvenile record	14.8%	12.0% (0.690)	28.6% (0.113)	0.0% (0.016)
felony charge	65.8%	58.0% (0.292)	81.0% (0.170)	41.4% (0.009)

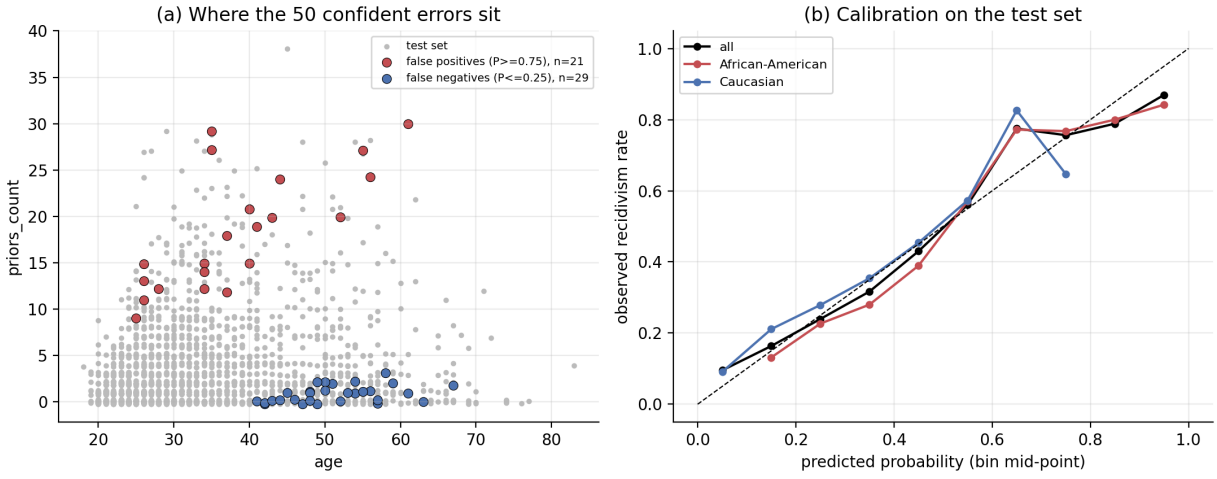


Figure 4. (a) Age and prior record of the 50 confident errors against the test set; (b) calibration of Model B on the test set, overall and by race (bins with fewer than 10 people omitted).

8 Part 2.4 – Fairness trade-off analysis

Groups. Sensitive variable: race. Group A = Caucasian (n = 747), Group B = African-American (n = 1116) in the test set (other race groups are excluded here). Test-set base rates are 41.1% (A) and 51.5% (B); mean Model B probabilities are 0.39 and 0.53. **Orientation.** The EO formula takes the *minimum* of two

ratios. With the more-flagged group in the denominator both ratios are ≤ 1 and the minimum reports the *larger* disparity; with the reverse orientation the minimum would report the smaller one and understate the problem. This orientation also keeps all points inside the required 0–1.5 axes. The reciprocal (A = African-American) table is in Appendix A. **Zero denominators.** A ratio is NA if its denominator is zero or a needed rate is undefined. This occurs once, at t = 0.95: nobody in Group A is flagged, so PP is NA (DP and EO are computed as 0 because only the numerator is zero). At that threshold only 11 Group B members are flagged, so those values carry almost no information.

Definitions (as in the brief). DP = $P(\hat{Y} = 1 \mid A) / P(\hat{Y} = 1 \mid B)$; EO = $\min(TPR_A/TPR_B, FPR_A/FPR_B)$; PP = $P(Y = 1 \mid \hat{Y} = 1, A) / P(Y = 1 \mid \hat{Y} = 1, B)$. $\hat{Y} = 1$ if $P \geq t$, for t = 0.05, 0.15, …, 0.95. Ideal value 1.0 for all three.

Table 13. Fairness metrics for every threshold (A = Caucasian, B = African-American; test set). EO is the smaller of the TPR and FPR ratios. NA: PP is undefined at t = 0.95 because no one in Group A is flagged. The recommended threshold t = 0.45 is highlighted.

t	Flagged A / B	DP	TPR ratio	FPR ratio	EO	PP	Accuracy
0.05	747 / 1116	1.000	1.000	1.000	1.000	0.798	45.4%
0.15	707 / 1102	0.958	0.976	0.950	0.950	0.812	48.1%
0.25	565 / 1048	0.805	0.890	0.755	0.755	0.882	55.6%
0.35	426 / 939	0.678	0.767	0.630	0.630	0.902	60.9%
0.45	252 / 742	0.507	0.585	0.478	0.478	0.920	66.8%
0.55	115 / 441	0.390	0.478	0.340	0.340	0.979	67.4%
0.65	41 / 258	0.237	0.306	0.176	0.176	1.028	63.0%
0.75	24 / 146	0.246	0.304	0.212	0.212	0.988	59.7%
0.85	7 / 68	0.154	0.204	0.095	0.095	1.060	57.0%
0.95	0 / 11	0.000	0.000	0.000	0.000	NA	54.9%

Table 14. Underlying group rates and error counts (Group A: 747 defendants, of whom 440 did not reoffend; Group B: 1116, of whom 541 did not reoffend).

t	TPR A	TPR B	FPR A	FPR B	PPV A	PPV B	FP A	FP B	FN A	FN B
0.05	100.0%	100.0%	100.0%	100.0%	41.1%	51.5%	440	541	0	0
0.15	97.4%	99.8%	92.7%	97.6%	42.3%	52.1%	408	528	8	1
0.25	87.6%	98.4%	67.3%	89.1%	47.6%	54.0%	296	482	38	9
0.35	71.3%	93.0%	47.0%	74.7%	51.4%	57.0%	207	404	88	40
0.45	48.5%	83.0%	23.4%	49.0%	59.1%	64.3%	103	265	158	98
0.55	27.4%	57.2%	7.0%	20.7%	73.0%	74.6%	31	112	223	246
0.65	10.7%	35.1%	1.8%	10.4%	80.5%	78.3%	8	56	274	373
0.75	6.2%	20.3%	1.1%	5.4%	79.2%	80.1%	5	29	288	458
0.85	2.0%	9.6%	0.2%	2.4%	85.7%	80.9%	1	13	301	520
0.95	0.0%	1.6%	0.0%	0.4%	NA	81.8%	0	2	307	566

Table 15. Percentile bootstrap intervals (1,000 resamples of the test set, seed 450). “valid” = share of resamples in which the ratio was estimable.

t	DP: 95% bootstrap CI	EO: 95% bootstrap CI	PP: 95% bootstrap CI
0.05	[1.00, 1.00]	[1.00, 1.00]	[0.72, 0.88]
0.15	[0.94, 0.98]	[0.92, 0.97]	[0.73, 0.90]
0.25	[0.77, 0.84]	[0.70, 0.81]	[0.79, 0.98]
0.35	[0.63, 0.73]	[0.57, 0.70]	[0.81, 1.00]
0.45	[0.46, 0.56]	[0.40, 0.56]	[0.82, 1.04]
0.55	[0.32, 0.46]	[0.23, 0.47]	[0.86, 1.10]
0.65	[0.17, 0.32]	[0.06, 0.30]	[0.86, 1.20]
0.75	[0.16, 0.36]	[0.04, 0.35]	[0.75, 1.22]
0.85	[0.05, 0.28]	[0.00, 0.24]	[0.60, 1.36] (100% of draws valid)
0.95	[0.00, 0.00]	[0.00, 0.00] (86% of draws valid)	NA (0% of draws valid)

Fig. 5 DP vs EO across thresholds (labels = threshold t)

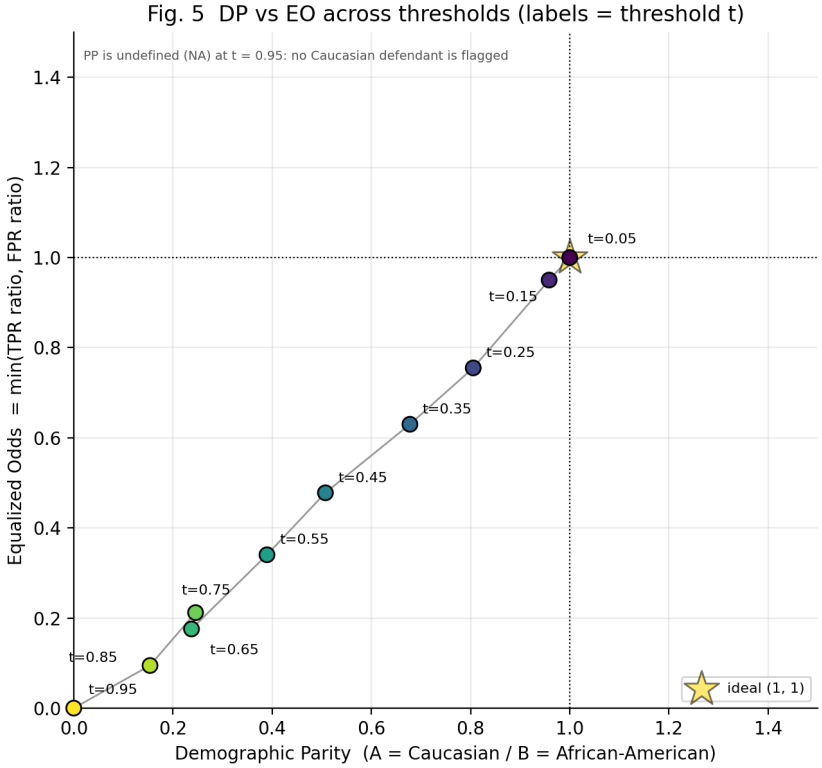


Figure 5. Required trade-off chart: Demographic Parity (x) against Equalized Odds (y), both on 0–1.5, each point labelled by its threshold. The star marks the ideal (1, 1).

Parity score. To compare gains and losses across metrics that can lie on either side of 1, each ratio m is converted to s = min(m, 1/m) (1 = perfect parity); percentages below are relative changes in s. **DP and EO do not trade off here.** Across the sweep they are almost perfectly correlated (r = 0.998), and no one-step reduction in threshold improves DP while worsening EO (0 of 9 steps), because EO's binding term is the false-positive-rate ratio, which tracks the flag-rate ratio. The assignment's template therefore cannot be completed with EO as the loser; the costs of parity land on predictive parity and accuracy. **Quantified trade-off.** If you improve DP from 0.390 (t = 0.55, the accuracy-maximising threshold) to 0.507 (t = 0.45), a 30.2% gain, EO also rises from 0.340 to 0.478 (+40.4%), but PP moves away from parity from 0.979 to 0.920 (a 6.1% loss) and accuracy falls 0.65 points. The trade-off cost is 6.1/30.2 = **0.20** percent of PP per percent of DP. One step further (t = 0.45 → 0.35) buys DP +33.6% and EO +31.8% for a PP loss of only 1.9% (cost 0.06) but costs 5.9 accuracy points (66.8% → 60.9%) and raises the number of wrongly flagged non-reoffenders in the two groups from 368 to 611. Pushing to t = 0.25 doubles DP (+107%) at an accuracy cost of 11.8 points. The ratios reach 1.0 only at t = 0.05, where everyone is flagged (accuracy 45.4%, below the 54.6% majority baseline): perfect parity is available only by abandoning prediction. **Key insight:** no threshold satisfies all three metrics. PP is within 3% of 1 only at t = 0.55, 0.65 and 0.75, and at those thresholds EO is at most 0.34.

Fig. 6 Fairness ratios and accuracy by threshold

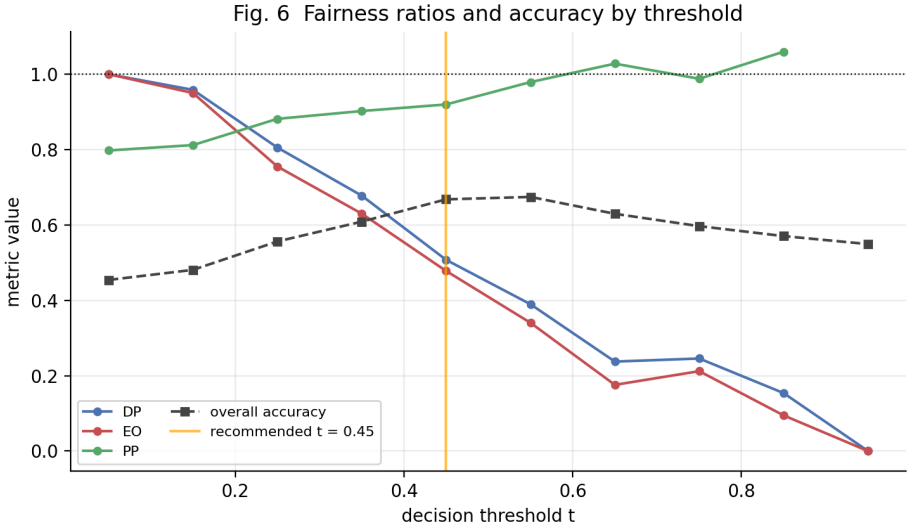


Figure 6. DP, EO, PP and overall accuracy across thresholds; the vertical line marks the recommended threshold.

Table 16. Quantified trade-offs between selected thresholds, computed on the parity score s = min(m, 1/m). All 9 consecutive steps are in results/tables/part24_consecutive_tradeoffs.csv.

Move (t)	DP	EO	PP	DP gain	EO gain	PP change	Accuracy change	Cost (PP% per DP%)
0.55 → 0.45	0.39 → 0.51	0.34 → 0.48	0.98 → 0.92	+30.2%	+40.4%	−6.1%	−0.6 pts	0.200
0.45 → 0.35	0.51 → 0.68	0.48 → 0.63	0.92 → 0.90	+33.6%	+31.8%	−1.9%	−5.9 pts	0.057
0.55 → 0.25	0.39 → 0.81	0.34 → 0.76	0.98 → 0.88	+106.7%	+121.9%	−10.0%	−11.8 pts	0.093
0.55 → 0.05	0.39 → 1.00	0.34 → 1.00	0.98 → 0.80	+156.7%	+193.8%	−18.5%	−22.0 pts	0.118

Decision rule and recommendation. Among thresholds whose accuracy is within 2 percentage points of the best (67.4% at t = 0.55), I choose the one with the highest worst-case parity score, min(s_{DP}, s_{EO}, s_{PP}). The rule selects **t = 0.45**, and the choice is unchanged for tolerances of 1 to 5 points but moves to t = 0.35 at 7 to 10 points, so the 2-point tolerance is a stated value judgment, not a data-driven fact. At t = 0.45: accuracy 66.8%; DP 0.507 (95% bootstrap CI [0.456, 0.562]); EO 0.478 [0.395, 0.562]; PP 0.920 [0.819, 1.036]. PP is statistically indistinguishable from parity, but DP and EO are not: the false-positive rate is 49.0% for African-American versus 23.4% for Caucasian defendants, and 265 versus 103 non-reoffenders are wrongly flagged. I recommend this threshold only for advisory use, paired with the monitoring described in Part 4.

9 Part 3 – Descriptive analysis questions

Each answer is written from the numbers computed above. Word counts are shown in grey (assignment target: about 250–300 words).

Question 1

Why did the threshold-based approach and the direct logistic approach flag, or not flag, the same individuals as high risk? Quote the mismatch count and percentage, and compare the median linear prediction with the 0.5 threshold.

The two rules disagreed on 296 of 2,165 test defendants (13.67%) and agreed on 1,869 (86.33%). The disagreement is nearly one-directional: 283 people were high risk under Model A but low risk under Model B, against only 13 the other way. Three mechanisms explain it. First, the cut-offs mean different things. Model A flags predictions at or above the median of its own output, 4.592 on COMPAS's 1–10 scale, so it flags 50.07% of defendants by construction. Model B's 0.5 is an absolute probability; because only 45.4% of test defendants reoffended, it flags 37.60%. Second, the targets differ. Model A imitates COMPAS's proprietary score and adds 0.587 decile points for African-American defendants ($p < 0.001$), whereas Model B, trained on real outcomes, finds a log-odds gap of only 0.098 ($p = 0.158$). Third, a linear model is unbounded: Model A predicts from −1.48 to 15.67. The median rule also changes the error profile: Model A's flag has higher recall (67.7% versus 56.1%) but lower precision (61.3% versus 67.7%) and a false-positive rate of 35.4% against 22.3%. Rankings still agree strongly (Spearman 0.960), so mismatches cluster near the boundary: their mean Model B probability is 0.457 (range 0.358–0.566). Cutting Model B at 0.449, which flags the same share as Model A, reduces mismatches to 220 (10.16%); so 76 disagreements (26%) are pure threshold semantics and the rest reflects genuinely different models.

Agreement is not accuracy: 568 defendants (26.2%) receive identical but wrong flags. Against actual outcomes Model A's flag is 66.0% accurate and Model B's 67.9%; on the disagreements Model B was right 169 times, Model A 127.

(267 words)

Question 2

Describe the subgroup with significantly different effects. Using the coefficients from the full model and both subgroups, explain why the trend differs, with exact coefficients and confidence intervals, and say which policy conclusion would change if the subgroup were ignored.

I split by sex (A = male, n = 5,819; B = female, n = 1,395). No strict Simpson-style reversal exists: of 26 scanned subgroup comparisons, 5 changed sign, but in none were both coefficients significantly non-zero. The robust finding is a stronger prior-record effect for women. Per standard deviation of priors_count the logit coefficient is 0.721 in the full data (odds ratio 2.06), 0.666 [0.594, 0.738] for men (OR 1.95) and 1.024 [0.813, 1.235] for women (OR 2.78). The intervals do not overlap; the difference test gives $z = 3.15$, $p = 0.0016$, and $p = 0.043$ after Bonferroni correction over 26 tests. Why? Much of the gap is functional form rather than a different mechanism. Women have fewer priors (mean 2.28, median 1, versus 3.76 and 2), so they sit on the steep part of a concave risk curve; with log(1 + priors) the coefficients converge to 0.666 versus 0.811 ($p = 0.075$). The pooled model's test calibration shows the consequence: women with at least five priors (n = 66) had predicted risk 0.582 against 0.667 observed, and women with at most one prior (n = 249) 0.288 against 0.245. If the subgroup were ignored, the pooled coefficient understates priors for women by about 30%, so a rule such as "flag five or more priors" would under-prioritise high-prior women and over-flag low-prior women. However, sex-specific models did not improve out-of-sample discrimination for women (AUC 0.718 versus 0.724 pooled), so the policy change is to audit calibration by sex, not to build separate scorecards. Splitting by race gives priors coefficients of 0.664 (African-American) versus 0.856 (Caucasian), nominal $p = 0.021$ but Bonferroni $p = 0.55$, so I do not claim it.

(282 words)

Question 3

State the variable dropped for the omitted-variable test. Report the before/after coefficient shift for at least four predictors in a table with percentage changes. Which coefficient shifted most, and what does that reveal about trusting an un-stress-tested model?

I dropped **priors_count**, the strongest predictor ($\beta = 0.701$) and one correlated with race_African_American ($r = 0.19$), juv_misd_count (0.22) and age (0.17). Table 17 shows the before/after coefficients, with $\%\Delta = (\beta_{reduced} - \beta_{full})/|\beta_{full}| \times 100$. By percentage, juv_misd_count moved most (−0.018 → 0.109, +708%, a sign flip), but its baseline is almost zero, so that figure is a denominator artefact. Measured in absolute terms and in standard errors, age shifted most (−0.526 → −0.312, +40.6%, 5.9 SE). The most consequential shift is race_African_American, from 0.098 ($p = 0.158$) to 0.302 ($p < 0.001$), +207.7%: the odds ratio rises from 1.10 to 1.35. Other shifts: juv_fel_count 0.054 → 0.184 (+241%), felony +70.6% and male +35.8%. The direction follows the omitted-variable logic: priors correlate positively with age and raise risk, so removing them weakens age's protective coefficient. Four coefficients became significant and none lost significance, and test AUC fell from 0.734 to 0.664. The lesson is that a coefficient is conditional on the other variables in the model: the reduced model would tell a policy-maker that race predicts recidivism after adjustment, while the full model finds no significant race effect. Both models use the same 5,049 training rows, so the shifts are caused by the omission alone. Two caveats apply: logistic coefficients are non-collapsible, so some change occurs even without confounding, and priors may be a mediator of policing exposure, so neither model is causal. A model whose coefficients have never been stress-tested this way is a snapshot of correlations that can rearrange when one variable is absent, so I would not read any single coefficient, least of all a race coefficient, at face value.

(273 words)

Table 17. Coefficient shifts after dropping priors_count (selected predictors; full table in Section 6).

Predictor	β full	β reduced (no priors_count)	% change	Significance change
age	−0.526	−0.312	+40.6%	unchanged
race_African_American	0.098	0.302	+207.7%	n.s. → significant
juv_fel_count	0.054	0.184	+241.3%	n.s. → significant
juv_misd_count	−0.018	0.109	+708.4%	n.s. → significant
juv_other_count	0.049	0.104	+111.4%	n.s. → significant
felony	0.208	0.354	+70.6%	unchanged
male	0.319	0.434	+35.8%	unchanged

Question 4

As a practitioner, how would you detect a hidden confound you cannot currently see? Propose one concrete, real-world method.

As a practitioner I would run a **covariate-augmentation and sensitivity audit** rather than trust the current feature set. Step 1: convene a domain-expert audit, with public defenders, probation officers and a county data steward, to list factors that plausibly drive both exposure to the justice system and reoffending but are absent from the file: employment, housing stability, substance-use treatment, supervision conditions and neighbourhood policing intensity. Step 2: obtain proxies for a pilot sample, for example census-tract arrest rates and unemployment joined by address, collected under data-protection approval. Step 3: refit the model with each candidate and measure coefficient shifts in standard-error units. My own experiment supplies the yardstick: omitting priors_count moved age by 5.9 SE and race_African_American by 2.9 SE. I would pre-register a trigger, for example any key or protected-attribute coefficient moving by more than 1 SE, which would send the model back for redesign. Step 4: for confounders that cannot be measured, quantify robustness with an omitted-variable sensitivity analysis (Cinelli and Hazlett, 2020) or an E-value (VanderWeele and Ding, 2017), asking how strong an unmeasured variable would have to be, relative to priors_count, to erase the effect of interest. Step 5: validate externally on another county or year and compare calibration within groups. Limitations are real: experts only name confounders they can imagine, proxies are noisy, and the audit costs time and data access. It is also unable to prove absence of confounding; it can only bound it. I would still prefer it to a single hold-out accuracy figure, which in this dataset (67.9%) said nothing about the 4 coefficients that changed significance when one variable disappeared.

(269 words)

Question 5

Describe the 50-row adversarial subset: the common characteristic, the percentage that share it, and the statistical reason the model fails confidently, referring to coefficients or decision boundaries.

The 50-row subset is 21 false positives and 29 false negatives, and the shared trait is **age together with prior record at the extremes**. None of the 50 is under 25 (test set: 20.0%); 56% are 45 or older (test set 21.2%). Within the subset, **100% of the false positives have five or more priors** (median 18 versus 2 in the test set), while **76% of the false negatives have at most one prior** and 83% are 45 or older; none has a juvenile record. The mechanism is linear extrapolation. In Model B each standard deviation of priors adds 0.701 log-odds and each of age subtracts 0.526. The false positives average +3.08 SD of priors, a contribution of +2.16 log-odds, so they sit at mean predicted probability 0.86; the false negatives average +1.38 SD of age (−0.72 log-odds) and −0.54 SD of priors, giving 0.19. Both sit in tails where the straight-line assumption is weakest: the extended model finds priors² = −0.086 (risk saturates) and age² = +0.196 (old age protects less than linear). Yet non-linearity is not the main story. Calibration shows 90 confident errors against 88.4 expected from the model's own probabilities, and the extended model raises test AUC only from 0.734 to 0.739 while making 114 confident errors. Most of these failures are outcome noise that no feature in the file can predict. Fisher exact tests against the remaining test rows confirm both patterns (priors ≥ 5: 100% versus 26.2%; age ≥ 45: 82.8% versus 20.3%; both $p < 0.001$). The errors also have a group pattern: 81% of the false positives are African-American, whereas 72% of the false negatives are Caucasian.

(275 words)

Question 6

Using the fairness trade-off chart, explain in concrete terms what the trade-off cost means for a real person. Describe two individuals, one from each group, and their outcomes under different thresholds.

Consider two people who would *not* reoffend: Ms A (Caucasian) and Mr B (African-American). Using the sweep in Figure 5, the chance that each is wrongly labelled high risk is 7.0% versus 20.7% at t = 0.55, 23.4% versus 49.0% at t = 0.45, 47.0% versus 74.7% at t = 0.35 and 67.3% versus 89.1% at t = 0.25. The sweep's EO (the FPR ratio) improves from 0.34 to 0.76, but only because Ms A's false-positive chance rises almost tenfold; parity is bought by flagging more innocent people, 778 across both groups at t = 0.25 against 143 at t = 0.55. Prioritising precision and accuracy (t = 0.55, PP 0.98) has the opposite cost: Mr B's chance of an unwarranted flag is 2.9 times Ms A's, while 72.6% of Caucasian reoffenders go unflagged against 42.8% of African-American reoffenders. At t = 0.45 that means 265 of 541 African-American non-reoffenders and 103 of 440 Caucasian non-reoffenders were wrongly flagged. Identical records are treated almost identically: a hypothetical 30-year-old man with three priors and a felony charge receives P = 0.53 if African-American and 0.51 if Caucasian, so both are flagged at t = 0.45 and neither at 0.55. The group gap therefore arises from different record distributions (mean priors 4.66 for Group B versus 2.45 for Group A; mean age 32.5 versus 37.4), which may themselves reflect unequal policing. No threshold removes the harm; each moves it between people. A judge reading a flag should know that, at t = 0.45, a non-reoffending African-American defendant faces a 49.0% chance of that label.

(261 words)

Question 7

Choose a threshold and argue, with at least three specific numbers, why it is defensible; give the strongest counterargument and acknowledge its validity.

I choose **t = 0.45**. The defence rests on six numbers. Accuracy is 66.8% (66.5% within the two groups analysed), only 0.65 points below the best (67.4% at t = 0.55) and 12.2 points above the 54.6% majority baseline. Relative to t = 0.55, DP improves from 0.390 to 0.507 (+30.2%) and EO from 0.340 to 0.478 (+40.4%) for that small accuracy cost. PP is 0.920 with a bootstrap interval [0.819, 1.036] that contains 1, so a "high-risk" label is about equally likely to be correct for both groups (59.1% versus 64.3%). The choice follows a stated rule: the highest worst-case parity score among thresholds within two accuracy points of the best, and it is stable for tolerances of one to five points. Lower thresholds look fairer on ratios but flag 84.1% of Group B at t = 0.35 and 93.9% at 0.25. **Strongest counterargument:** EO = 0.478 means non-reoffending African-American defendants are flagged 2.09 times as often (49.0% versus 23.4%; 265 versus 103 people), and the intervals for DP [0.456, 0.562] and EO [0.395, 0.562] exclude 1. A critic could say a threshold with that disparity is not fair in any practical sense, and could point to t = 0.35 (EO 0.630). Fairness also cuts the other way here: at this threshold the model catches 83.0% of African-American reoffenders but only 48.5% of Caucasian reoffenders. I accept the objection's validity. t = 0.35 does improve EO, but at an accuracy cost of 5.9 points and with 611 wrongly flagged people instead of 368. My choice is a values judgment about that exchange rate, not a fact the data can settle.

(271 words)

Question 8

Should a policymaker prioritise demographic parity or equalized odds for this dataset? Cite the trade-off numbers and acknowledge what is sacrificed.

I would recommend prioritising **equalized odds**, specifically parity in false-positive rates, over demographic parity. The groups' base rates differ (51.5% versus 41.1% in the test set), so demographic parity would require equal flag rates regardless of what people did. At t = 0.45 the flag rates are 66.5% (B) and 33.7% (A), and closing that gap by lowering the threshold to 0.25 (DP 0.507 → 0.805, +59%) means flagging 93.9% of Group B and 75.6% of Group A: mass flagging, with overall accuracy falling to 55.6%. Equalized odds conditions on the actual outcome, so it targets the concrete harm in this setting: non-reoffenders being wrongly detained or denied release, a direct loss of liberty whose burden can be audited from outcomes alone. The false-positive ratio is 0.478 at t = 0.45 and 0.755 at t = 0.25. In this model the two metrics rise together (r = 0.998; EO +58% over the same move), so choosing EO does not sacrifice DP; what is sacrificed is predictive parity and accuracy. PP slips from 0.920 to 0.882 (4.2% lower) and accuracy by 11.2 points. I find that acceptable because PP stays within 12% of parity, whereas a 0.48 false-positive ratio is not. Two cautions: EO, defined as a minimum of ratios, can hide a large gap in the weaker ratio; and a race-blind model still gives DP 0.570 and EO 0.573 at t = 0.45, so the disparity runs through proxies such as priors and age, and policy must also scrutinise those inputs. I would write this into policy as a published ceiling on the false-positive ratio, audited annually.

(267 words)

10 Part 4 – Reflection essay

"If a policy-maker asked you: 'just give me one number: should we deploy this model or not?', what would you say, and what is the limitation of that 'one number' answer that you discovered yourself through your analysis?"

Direct answer. I would recommend **conditional deployment**: advisory decision support only, at t = 0.45, with mandatory group-level monitoring, and not as an automated trigger for detention or sentencing. The headline reason is that 66.8% overall accuracy conceals a false-positive rate of 49.0% for African-American versus 23.4% for Caucasian defendants.

Why a single number fails. First, *overall accuracy hides subgroup gaps*. The headline 67.9% at 0.5 says nothing about who bears the errors: at t = 0.45 the false-positive ratio is 0.48 and the true-positive ratio 0.59, and even discrimination differs by group (test AUC 0.748 for African-American versus 0.689 for Caucasian defendants); the pooled model also under-predicts Caucasian risk (0.385 predicted versus 0.411 observed). Second, *the threshold is a values question, not a data-driven fact*. Accuracy peaks at t = 0.55 (67.4%); the ratio metrics are "perfect" only at t = 0.05, where everyone is flagged and accuracy (45.4%) falls below the 54.6% baseline. My own rule picks 0.45 for any accuracy tolerance from one to five points and 0.35 from seven to ten, so the answer depends on a human choice about how much accuracy to trade. Even the conventional 0.5 is a choice: it misses 43.9% of reoffenders. Third, *fairness metrics conflict*. Base rates differ (51.5% versus 41.1%), so predictive parity (0.98 at t = 0.55) and equalized odds (0.34 at the same threshold) cannot both hold; improving EO to 0.76 at t = 0.25 flags 778 non-reoffenders instead of 143. Fourth, *the model fails confidently on predictable subsets*: 90 test cases (4.2%) are confident errors, concentrated among defendants with five or more priors (false positives) and older defendants with almost none (false negatives), and a single accuracy number gives no hint of where. Fifth, *hidden confounds remain unknowable*: omitting one variable moved the race coefficient from 0.098 to 0.302 and made 4 coefficients significant, so any variable I have not measured could do the same. Finally, *every ratio carries sampling error*: with 747 Caucasian test defendants, DP at t = 0.45 has a 95% bootstrap interval of [0.456, 0.562] and PP [0.819, 1.036], which cannot distinguish parity from a gap of roughly 18%. A policy-maker given one number would inherit all of these problems without being able to see them.

Grounding in evidence. Part 1: 296 mismatched flags (13.67%), median 4.592 versus the 0.5 threshold. Part 2.1: priors effect 1.024 (women) versus 0.666 (men), p = 0.0016. Part 2.2: race coefficient 0.098 → 0.302. Part 2.3: 90 confident errors against 88.4 expected (21 false positives and 29 false negatives in the subset). Part 2.4 at t = 0.45: DP 0.507, EO 0.478, PP 0.920, accuracy 66.8%.

Closing. The meta-lesson is that responsible deployment means publishing the trade-offs, not hiding them inside one metric. Decision-makers should see the threshold, the error rates by group, the confident failures and the unresolved confounds, and be able to challenge each. A model card carrying these tables, rather than a score alone, is what I would hand to the policy-maker.

(499 words)

11 Limitations and conclusion

Limitations. (1) The target is two-year re-arrest recorded in one Florida county, which measures policing as well as behaviour; label bias cannot be tested with this file. (2) The analysis uses one random split (seed 450); the fairness estimates for Group A rest on 747 test defendants, so bootstrap intervals are wide (for example DP at t = 0.45: [0.456, 0.562]) and values at t ≥ 0.85 rest on very few flagged people. (3) Section 2.1 was a search, so I applied a Bonferroni correction across 26 comparisons and selected the reported split after the scan; the selection is declared, not hidden. (4) Race was included as a predictor to expose its effect; removing it did not remove the disparity (Appendix B). (5) Logistic coefficients are non-collapsible and priors may be a mediator, so no causal claim is made. (6) The recommended threshold depends on a stated 2-point accuracy tolerance. (7) The EO definition (a minimum of ratios) can mask the weaker disparity; Table 13 reports both ratios. **Conclusion.** A simple, transparent logistic model predicts two-year recidivism with accuracy 67.9% and AUC 0.734, yet every stress test exposed something the headline metrics hide: disagreement with the linear score (13.67%), coefficient fragility, structured confident errors and unequal error rates across groups. The defensible conclusion is conditional, advisory use with explicit, published trade-offs.

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Appendix A – Fairness sweep with the reciprocal orientation

The same classifier and test set with Group A = African-American and Group B = Caucasian. Every ratio is the reciprocal of the main table; EO now equals the *smaller* of the two ratios, which understates the larger disparity (for example at t = 0.45 the FPR ratio is 2.092 but EO reports 1.709). This is why the main analysis uses the opposite orientation.

Table 18. Reciprocal orientation (A = African-American, B = Caucasian). NA at t = 0.95: Group B (Caucasian) has no flagged defendants, so every ratio has a zero denominator.

t	DP	TPR ratio	FPR ratio	EO	PP
0.05	1.000	1.000	1.000	1.000	1.254
0.15	1.043	1.025	1.053	1.025	1.232
0.25	1.242	1.123	1.324	1.123	1.134
0.35	1.475	1.304	1.587	1.304	1.108
0.45	1.971	1.709	2.092	1.709	1.087
0.55	2.567	2.091	2.938	2.091	1.021
0.65	4.212	3.268	5.693	3.268	0.973
0.75	4.072	3.288	4.717	3.288	1.012
0.85	6.502	4.894	10.573	4.894	0.944
0.95	NA	NA	NA	NA	NA

Appendix B – Robustness checks

Model B was refitted under three alternative choices: the alternative outcome *is_recid*; winsorised priors_count; and a "race-blind" model without race indicators. Fairness metrics are shown at the recommended threshold (A = Caucasian, B = African-American). The race-blind model keeps its AUC but does not eliminate the disparity, which runs through proxies such as prior record and age.

Table 19. Robustness checks for Model B.

Variant	AUC	Accuracy at 0.5	Brier	DP at 0.45	EO at 0.45	PP at 0.45
Main model B (target two_year_recid)	0.734	0.679	0.208	0.507	0.478	0.920
Target = is_recid	0.740	0.680	0.208	0.544	0.519	0.886
Priors winsorised at train p99 (22)	0.735	0.678	0.208	0.509	0.478	0.921
Race-blind (no race indicators)	0.735	0.684	0.208	0.570	0.573	0.886

Appendix C – Reproducibility and quality control

How to regenerate. Install the packages in requirements.txt, open notebook/CSET485_Assignment1_COMPAS.ipynb and run all cells. The notebook downloads the dataset if data/compas-scores-two-years.csv is absent, fits every model with random_state = 450, and writes results/results.json, results/tables/*.csv and results/figures/*.png. Then run *python src/build_report.py* to rebuild this PDF (the last notebook cell does this automatically when reportlab is installed). **Environment used:** Python 3.12.3, numpy 2.4.4, pandas 3.0.2, scikit-learn 1.8.0, statsmodels 0.15.0, scipy 1.17.1, matplotlib 3.10.8. **Data file SHA-256:** c451db85908b2f7fef1d83203bedf6b71ecda0d5af468d82ae62178f91d0cc7d.

Checks performed. The first block is evaluated when this PDF is built; the second block was run once as an independent re-computation (src/verify_results.py) that refits the models with a different library and compares headline numbers.

Table 20. Quality-control checklist.

Check	Result
Dataset is COMPAS two-year file	PASS
Random seed is 450	PASS
Split is 70/30	PASS
Models A (OLS) and B (logistic) implemented and cross-checked with scikit-learn	PASS
Four stress tests present (2.1–2.4)	PASS
Fairness sweep covers all 10 thresholds 0.05–0.95	PASS
All eight Part 3 answers within 250–300 words	PASS
Reflection about 500 words with the four required parts	PASS
Mismatch count equals off-diagonal of the confusion matrix	PASS
Adversarial subset: 50 rows drawn with seed 450	PASS
Independent refit: median of Model A predictions	PASS (4.5923)
Independent refit: mismatch count between the two flags	PASS (296)
Independent refit: Model B accuracy and AUC	PASS (acc 0.6790, AUC 0.7341)
Independent refit: confident false positives / false negatives	PASS (35 / 55)
Independent recomputation of DP / EO / PP at all 10 thresholds	PASS (max abs difference 4.8e-07; NA pattern identical)
Independent refit: reduced-model race coefficient	PASS (0.3024 vs 0.3024)
Independent refit: priors_count coefficient, men	PASS (0.6660 vs 0.6660)
Independent refit: priors_count coefficient, women	PASS (1.0240 vs 1.0240)